

## Performance Testing

|                      |                                                  |
|----------------------|--------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                    |
| <b>Team ID</b>       | xxxxxx                                           |
| <b>Project Name</b>  | Smart agriculture production optimization engine |
| <b>Maximum Marks</b> | 5 Marks                                          |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                      |
|----------------------------|----------------------------------------------|
| <b>Testing Tool Used</b>   | JMeter, Locust, k6, Postman                  |
| <b>Type of Testing</b>     | Load Testing, Stress Testing, Spike Testing  |
| <b>Target Module / API</b> | Specify which feature or endpoint was tested |
| <b>Test Environment</b>    | Local / Cloud / Hosted Server                |
| <b>Test Date</b>           |                                              |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome |
|------|-----------------------------|----------------------|----------------|------------------|
| 1    |                             |                      |                |                  |
| 2    |                             |                      |                |                  |
| 3    |                             |                      |                |                  |
| 4    |                             |                      |                |                  |

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks |
|------|----------------------|--------------|--------------|----------------------|---------|
| 1    | Response Time (Avg)  | < 2 seconds  |              |                      |         |
| 2    | Response Time (Max)  | < 5 seconds  |              |                      |         |
| 3    | Throughput (Req/sec) |              |              |                      |         |
| 4    | Error Rate           | < 1%         |              |                      |         |

5 CPU Utilization < 80%

6 Memory Utilization < 80%

### Step 4: Observations & Analysis

#### Key Findings:

[Describe what the test results revealed about your system's performance]

#### Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

#### Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]

### Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]